Solution:

$y = f(x) = x^3 + 2x + 1$ for $x = 1,2,3,4,5$

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
1	4				
		9			
2	13		12		
		21		6	
3	34		18		0
		39		6	
4	73		24		
		63			
5	136				

Example 5.3

By constructing a difference table and using the second order differences as constant, find the sixth term of the series 8,12,19,29,42…

Solution:

Let k be the sixth term of the series in the difference table.

First we find the forward differences.

x	y	Δy	$\Delta^2 y$
1	8		
		4	
2	12		3
		7	
3	19		3
		10	
4	29		3
		13	
5	42		k–55
		k–42	
6	k		

Given that the second differences are constant

$\therefore \quad k - 55 = 3$

$k = 58$

$\therefore$ the sixth term of the series is 58

Example 5.4

Find (i) Δe^{ax} (ii) $\Delta^2 e^x$ (iii) $\Delta \log x$

Solution:

(i) $\Delta e^{ax} = e^{a(x+h)} - e^{ax}$

$= e^{ax}.e^{ah} - e^{ax} \quad \left[\because a^{m+n} = a^m.a^n\right]$

$= e^{ax}\left[e^{ah} - 1\right]$

(ii) $\Delta^2 e^x = \Delta.\left[\Delta e^x\right]$

$= \Delta\left[e^{x+h} - e^x\right]$

$= \Delta\left[e^x e^h - e^x\right]$

$= \Delta e^x\left[e^h - 1\right]$

$= \left(e^h - 1\right)\Delta e^x$

$= \left(e^h - 1\right).\left(e^h - 1\right).e^x$

$= \left(e^h - 1\right)^2.e^x$

(iii) $\Delta \log x = \log(x+h) - \log x$

$= \log\dfrac{x+h}{x}$

$= \log\left(\dfrac{x}{x} + \dfrac{h}{x}\right)$

$= \log\left(1 + \dfrac{h}{x}\right)$

Example 5.5

Evaluate $\Delta\left[\dfrac{5x+12}{x^2+5x+6}\right]$ by taking '1' as the interval of differencing.

Solution:

$\Delta\left[\dfrac{5x+12}{x^2+5x+6}\right]$

By Partial fraction method

$\dfrac{5x+12}{x^2+5x+6} = \dfrac{A}{x+3} + \dfrac{B}{x+2}$

$A = \dfrac{5x+12}{x+2}\left[x = -3\right]$

$= \dfrac{-15+12}{-1} = \dfrac{-3}{-1} = 3$

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